Grade 9 Practice

Standard A.SSE.1

Computation

1. In $3x + 7$, what is the coefficient of x? ___ **3**

2. In $2x^2 - 5$, what is the constant term? ___ **-5**

3. How many terms are in $4x - 9$? ___ **2**

4. In $5(x + 2)$, what is the factor outside the parentheses? ___ **5**

Word problems

5. The cost is $C = 12h + 30$, where h is hours. What does 12 represent?
Answer: the rate per hour

6. The value is $V = 500 - 25t$, where t is days. What does 25 represent?
Answer: the rate of decrease per day

Grade 9 Practice

Standard A.SSE.2

Computation

1. Rewrite $x^2 - 9$ as a product of two factors. ___ **$(x - 3)(x + 3)$**

2. Factor $x^2 + 5x + 6$. ___ **$(x + 2)(x + 3)$**

3. Factor $4x^2 - 1$. ___ **$(2x - 1)(2x + 1)$**

4. Rewrite $a^2 + 2ab + b^2$ as a square of a binomial. ___ **$(a + b)^2$**

Word problems

5. A square patio has area $x^2 + 6x + 9$. What is the side length?
Answer: $x + 3$

6. Factor $9 - x^2$.
Answer: $(3 - x)(3 + x)$

Grade 9 Practice

Standard A.APR.1

Computation

1. Simplify $(x + 3) + (2x - 1)$. ___ **$3x + 2$**

2. Multiply $(x + 2)(x + 5)$. ___ **$x^2 + 7x + 10$**

3. Simplify $(3x - 4) - (x + 6)$. ___ **$2x - 10$**

4. Multiply $2x(x - 3)$. ___ **$2x^2 - 6x$**

Word problems

5. A rectangle has sides $x + 2$ and $x + 4$. What is its area?
Answer: $x^2 + 6x + 8$

6. Add $(2x^2 + 3)$ and $(x^2 - x)$. What is the sum?
Answer: $3x^2 - x + 3$

Grade 9 Practice

Standard A.CED.1

Computation

1. Write an equation: 5 more than 3 times n is 20. ___ **3n + 5 = 20**

2. Write an equation: twice x minus 7 equals 11. ___ **2x − 7 = 11**

3. Write an inequality: y is at least 4. ___ **y ≥ 4**

4. Write an equation: x divided by 4 equals 6. ___ **x/4 = 6**

Word problems

5. A phone plan costs $15 plus $0.10 per text. Your bill is $25. Write an equation using t for texts.
Answer: 15 + 0.10t = 25

6. A team scored 3 times its points p plus 4 points, for a total of 31. Write an equation.
Answer: 3p + 4 = 31

Grade 9 Practice

Standard A.CED.2

Computation

1. In the line y = 2x + 1, what is the y-intercept? ___ **1**

2. In the line y = -x + 4, what is the slope? ___ **-1**

3. Is the point (2, 5) on the line y = 2x + 1? Write yes or no. **yes**

4. Is the point (0, 3) on the line y = x - 3? Write yes or no. **no**

Word problems

5. A cost is y = 5x + 10, where x is hours. What does the 10 represent?
Answer: a fixed cost of 10

6. A car's distance is y = 60x, where x is hours. What does the slope represent?
Answer: 60 miles per hour

Grade 9 Practice

Standard A.CED.3

Computation

1. Solve the system: x + y = 10 and x - y = 4. Write (x, y). **(7, 3)**

2. Solve the system: y = 2x and y = x + 3. Write (x, y). **(3, 6)**

3. Pens cost p and pencils cost c. 3p + 2c = 11 and p + 2c = 7. What is the cost of one pe

4. Adult tickets a and child tickets c: a + c = 50 and a - c = 10. What is a? ___ **30**

Word problems

5. Tickets: a + c = 100 and 8a + 5c = 620, where a is adult and c is child tickets. How many of each were sold?
Answer: 40 adult, 60 child

6. Nickels n and dimes d: n + d = 12 and 5n + 10d = 85. How many nickels and how many dimes?
Answer: 7 nickels, 5 dimes

Grade 9 Practice

Standard A.CED.4

Computation

1. Solve $y = mx + b$ for b. b = ___ **$y - mx$**

2. Solve $2x + y = 6$ for y. y = ___ **$6 - 2x$**

3. Solve $A = lw$ for w. w = ___ **A/l**

4. Solve $C = 2\pi r$ for r. r = ___ **$C/(2\pi)$**

Word problems

5. Distance $d = rt$. Solve for t.
Answer: $t = d/r$

6. Perimeter $P = 2l + 2w$. Solve for w.
Answer: $w = (P - 2l)/2$

Grade 9 Practice

Standard A.REI.1

Computation

1. Solve x/2 = 5. x = ___ **10**

2. Solve 4x - 8 = 0. x = ___ **2**

3. Solve 5 = x - 3. x = ___ **8**

4. Solve 3x = -9. x = ___ **-3**

Word problems

5. Sam says x = 2 solves 3x + 1 = 7. Is he right? Write yes or no.
Answer: yes

6. Mia solves 2x + 3 = 11 by subtracting 3 and then dividing by 2. Does she get x = 4? Write yes or no.
Answer: yes

Grade 9 Practice

Standard A.REI.3

Computation

1. Solve $x + 3 > 7$. $x >$ ___ **4**

2. Solve $2x \le 10$. $x \le$ ___ **5**

3. Solve $-3x < 9$. $x >$ ___ **-3**

4. Solve $4x - 5 \ge 11$. $x \ge$ ___ **4**

Word problems

5. Tickets cost $8 each. You can spend at most $64. Write an inequality for t tickets and solve it.
Answer: $t \le 8$

6. A game needs at least 30 points. You earn 4 points per round. What is the fewest whole number of rounds needed?
Answer: 8 rounds

Grade 9 Practice

Standard A.REI.4

Computation

1. Solve $x^2 = 16$. x = ___ **±4**

2. Solve $x^2 - 9 = 0$. x = ___ **±3**

3. Solve $(x - 2)^2 = 25$. x = ___ **7 or −3**

4. Solve $x^2 = 0$. x = ___ **0**

Word problems

5. A square has area 49 square units. What is its side length?
Answer: 7 units

6. A square has area 81 square units. What is its side length?
Answer: 9 units

Grade 9 Practice

Standard A.REI.5

Computation

1. Does multiplying $x + y = 4$ by 2 give $2x + 2y = 8$? Write yes or no. **yes**

2. Is $2x = 6$ equivalent to $x = 3$? Write yes or no. **yes**

3. Multiply $x + y = 4$ by 2. Write the new equation. **$2x + 2y = 8$**

4. Is $3x = 6$ equivalent to $6x = 12$? Write yes or no. **yes**

Word problems

5. Solve $2x - 4 = 6$ and $4x - 8 = 12$. Do they have the same solution? Write yes or no.
Answer: yes

6. If you multiply both sides of an equation by 0, do the solutions stay the same? Write yes or no.
Answer: no

Grade 9 Practice

Standard A.REI.6

Computation

1. Solve the system: y = x + 2 and y = -x. Write (x, y). **(-1, 1)**

2. Solve the system: y = 2x + 1 and y = 2x + 4. Write yes if there is a solution or no if there

3. Solve the system: x + y = 6 and x - y = 2. Write (x, y). **(4, 2)**

4. Solve the system: y = 3x and y = x + 4. Write (x, y). **(2, 6)**

Word problems

5. Plan A costs y = 5x + 10 and plan B costs y = 3x + 30. For what x do the costs match?
Answer: x = 10

6. Solve the system: x + y = 20 and y = 3x. Write (x, y).
Answer: (5, 15)

Grade 9 Practice

Standard F.IF.1

Computation

1. If $f(x) = 2x + 3$, what is $f(4)$? ___ **11**

2. If $g(x) = x^2$, what is $g(-3)$? ___ **9**

3. If $h(x) = x - 5$, what is $h(10)$? ___ **5**

4. If $f(x) = 7x - 2$, what is $f(0)$? ___ **-2**

Word problems

5. A phone cost is $c(t) = 10 + 5t$, where t is months. What is $c(3)$?
Answer: 25 dollars

6. If $f(x) = x^2 - 1$, which input gives $f(x) = 0$? Write your answer.
Answer: x = 1 or x = -1

Grade 9 Practice

Standard F.IF.2

Computation

1. If $f(x) = 3x + 1$, find $f(2)$. ___ **7**

2. If $g(x) = x^2 + 2x$, find $g(3)$. ___ **15**

3. If $h(x) = 10 - x$, find $h(-4)$. ___ **14**

4. If $f(x) = x/2$, find $f(8)$. ___ **4**

Word problems

5. The temperature is $T(h) = 30 - 2h$, where h is hours. What is $T(5)$?
Answer: 20 degrees

6. If $A(s) = s^2$, find $A(6)$.
Answer: 36

Grade 9 Practice

Standard F.IF.4

Computation

1. What is the y-intercept of y = 3x - 6? ___ **-6**

2. What is the x-intercept of y = 2x - 8? ___ **4**

3. What is the slope of y = -x + 2? ___ **-1**

4. Does the graph of y = 2x increase? Write yes or no. **yes**

Word problems

5. The graph of $y = x^2$ has its lowest point at (0, 0). Is that the vertex? Write yes or no.
Answer: yes

6. A temperature graph falls after noon. Is it increasing or decreasing after noon?
Answer: decreasing

Grade 9 Practice

Standard F.BF.1

Computation

1. A pile starts at 10 and adds 4 each day. Write the function f(x) for x days. **f(x) = 4x + 1**

2. A number is multiplied by 3. Write f(x). **f(x) = 3x**

3. A number is squared and then 1 is subtracted. Write f(x). **f(x) = x² - 1**

4. A number is doubled and then 5 is added. Write f(x). **f(x) = 2x + 5**

Word problems

5. A phone plan costs $30 plus $5 per month. Write a function f(m) for the cost after m months.
Answer: f(m) = 5m + 30

6. A bacteria count starts at 100 and doubles every hour. Write a function f(t) for t hours.
Answer: f(t) = 100 · 2^t

Grade 9 Practice

Standard F.LE.1

Computation

1. A table increases by 2 at each step. Is it linear or exponential? **linear**

2. A table doubles at each step. Is it linear or exponential? **exponential**

3. A table increases by 5 at each step. Is it linear or exponential? **linear**

4. A table multiplies by 3 at each step. Is it linear or exponential? **exponential**

Word problems

5. A population grows by 10% each year. Is it linear or exponential?
Answer: exponential

6. A salary increases by $500 each year. Is it linear or exponential?
Answer: linear

Grade 9 Practice

Standard F.LE.2

Computation

1. A line passes through (0, 3) and (1, 5). Write the equation. **y = 2x + 3**

2. An exponential function passes through (0, 2) and (1, 6). Write the equation. **y = 2 · 3**

3. A line has slope 4 and y-intercept -1. Write the equation. **y = 4x - 1**

4. An exponential function starts at 5 and doubles each step. Write the equation. **y = 5**

Word problems

5. Savings start at $100 and grow by $20 each month. Write an equation for y after x months.
Answer: y = 20x + 100

6. A value is cut in half each year, starting at 800. Write an equation for y after x years.
Answer: y = 800(1/2)^x

Grade 9 Practice

Standard S.ID.1

Computation

1. Find the range of 2, 4, 6, and 8. ___ **6**

2. Find the mean of 1, 2, 3, 4, and 5. ___ **3**

3. Find the median of 1, 3, 5, and 7. ___ **4**

4. Find the mode of 2, 2, 3, and 4. ___ **2**

Word problems

5. Five scores are 80, 85, 90, 95, and 100. What is the mean?
Answer: 90

6. A histogram has 6 values in the 10–20 bin and 4 in the 20–30 bin. How many values in all?
Answer: 10

Grade 9 Practice

Standard S.ID.2

Computation

1. Which measure of center is resistant to outliers? Write mean or median. **median**

2. Find the mean of the data 1, 2, 3, 4, and 5. ___ **3**

3. What does standard deviation measure? Write spread or center. **spread**

4. For symmetric data, how do the mean and median compare? Write equal or not equ

Word problems

5. Income data is skewed to the right. Which measure of center is better to use?
Answer: median

6. Data are 10, 12, 14, and 16. What is the mean?
Answer: 13